Daniel Köstler

A1:

$\sigma = 65{,}3 \frac{S}{m}$

$A = 1dm^2 = 0{,}01m^2$

$l = 0{,}008m$

$\rho = \frac{1}{\sigma}$

$R = \rho \cdot \frac{l}{A} = \frac{1}{\sigma} \cdot \frac{l}{A} = \frac{1}{65{,}3} \cdot \frac{0{,}008}{0{,}01} = 0{,}0123\,\Omega$

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A2:

$I = 40\,mA = 40 \cdot 10^{-3}\,A$

$R = 1\,k\Omega = 1000\,\Omega$

$I = \frac{U}{R} \Rightarrow U = R \cdot I$

$U = 40 \cdot 10^{-3} \cdot 10^{3} = 40\,V$

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A3:

$U = 15\,V$, $R_1 = 1000\,\Omega$, $R_2 = 4700\,\Omega$, $R_3 = 1000\,\Omega$

$$\frac{1}{R_{12}} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$R_{12} = \frac{R_1 \cdot R_2}{R_1 + R_2} = \frac{4700 \cdot 1000}{4700 + 1000} = 824{,}56\,\Omega$$

$$\frac{U_{12}}{U} = \frac{R_{12}}{R_{12} + R_3}$$

$$U_{12} = \frac{R_{12}}{R_{12} + R_3} \cdot U = \frac{824{,}56}{824{,}56 + 1000} \cdot 15 = 6{,}78\,V$$

$$U = U_{12} + U_3 \Longrightarrow U_3 = U - U_{12}$$

$$U_3 = 15 - 6{,}78 = 8{,}22\,V$$

$$I_3 = I = \frac{U_3}{R_3} = \frac{8{,}22}{1000} = 0{,}00822\,A = 8{,}22\,mA$$

$$I_2 = \frac{U_{12}}{R_2} = \frac{6{,}78}{4700} = 0{,}00144\,A = 1{,}44\,mA$$

$$I_1 = \frac{U_{12}}{R_1} = \frac{6{,}78}{1000} = 0{,}00678\,A = 6{,}78\,mA$$

$$U_1 = I_1 \cdot R_1 = 0{,}00678 \cdot 1000 = 6{,}78\,V$$
$$U_2 = I_2 \cdot R_2 = 0{,}00144 \cdot 4700 = 6{,}78\,V$$
$$U_3 = I_3 \cdot R_3 = 0{,}00822 \cdot 1000 = 8{,}22\,V$$

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A8:

$R = 330\,\Omega$

$U_0 = 10\,V$

$I = 25\,mA = 25 \cdot 10^{-3}\,A$

$$I = \frac{U_0}{R + R_i}$$

$$I \cdot (R + R_i) = U_0$$

$$R_i = \frac{U_0}{I} - R$$

$$R_i = \frac{10}{25 \cdot 10^{-3}} - 330$$

$$R_i = 70\,\Omega$$

$$I_{neu} = \frac{U_0}{R}$$

$$I_{neu} = \frac{10}{330} = \frac{1}{33}\,A$$

$$\Delta I = \frac{I_{neu} - I}{I} = \frac{7}{33} = 0{,}21\,A$$

$$\Delta I = 21\,\%$$

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A11:

$$I = \frac{U}{R}$$

$$I = \frac{U}{\rho \cdot \frac{L}{A}}$$

$$I = \frac{U \cdot A}{\rho \cdot L}$$

$$I \cdot \rho \cdot L = U \cdot A$$

$$L = \frac{U \cdot A}{I \cdot \rho}$$

$$L_{Alt} = \frac{U \cdot A}{I \cdot \rho}$$

$$L_{neu} = \frac{1}{0{,}7} \cdot \frac{U \cdot A}{I \cdot \rho}$$

$$L_{neu} - L_{Alt} = \frac{1}{0{,}7} \cdot \frac{U \cdot A}{I \cdot \rho} - \frac{U \cdot A}{I \cdot \rho} = \frac{U \cdot A}{I \cdot \rho} \cdot \left(\frac{1}{0{,}7} - 1\right)$$

$$= \frac{U \cdot A}{I \cdot \rho} \cdot 0{,}43$$

Die Leitung kann um ca. 43% verlängert werden.